

Statement of Purpose

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Artificial intelligence is advancing rapidly but not equally. The models powering modern NLP, healthcare diagnostics, and intelligent systems are built overwhelmingly on data from a small number of high-resource languages and well-funded institutions. The communities that lack large digital corpora, centralized data infrastructure, or institutional research capacity are systematically excluded from the benefits of the technology they increasingly depend on.

My research is motivated by this gap and by the conviction that closing it requires more than applying existing methods to new languages or domains. It requires rethinking how we train models, how we protect the data used to train them, and how we ensure that what they produce can be trusted.

Academic Foundation

I completed my B.Sc. in Computer Science and Engineering from the University of Asia Pacific, Bangladesh in 2024 with a CGPA of 3.87/4.00, and I am currently pursuing my M.Sc. in the same discipline with a perfect CGPA of 4.00/4.00. Both degrees were completed under active research supervision, with thesis work that directly contributed to peer-reviewed publications.

My academic record reflects consistent performance across all semesters. I received the Vice Chancellor's Honor Award five times awarded to students maintaining a CGPA of 3.90 or above in a semester and the Dean's Honor Award once. These recognitions were not incidental; they reflect a deliberate commitment to academic rigor alongside active research and extracurricular schedule.

Research Focus

I work at the intersection of **privacy-preserving machine learning** and **large language models for low-resource and underserved contexts**.

These two areas are more connected than they might appear. Low-resource communities, whether defined by language, geography, or institutional capacity are also the communities least able to absorb the risks of centralized AI: data exploitation, model unreliability, and opaque decision-making in high-stakes domains like healthcare and law. Building AI that is private, grounded, and explainable is not a technical nicety for these communities. It is a prerequisite for AI that is safe to use at all.

My work addresses this through two complementary directions:

Federated and privacy-preserving learning: designing training frameworks that allow models to learn from distributed, sensitive data without centralizing it. This is relevant wherever data governance is a constraint: hospitals that cannot share patient records, legal systems that cannot expose case data, IoT networks spanning heterogeneous and privacy-sensitive devices.

Reliable language models for low-resource contexts: developing and evaluating LLMs that remain factually grounded when deployed in domains and languages with sparse training data. Hallucination and factual drift are problems for all LLMs, but they are acute in low-resource settings where retrieval corpora are thin and ground-truth validation is difficult.

Research in Practice

My B.Sc. thesis demonstrated that federated learning could achieve competitive performance on a real-world classification task Bangla sign language recognition while eliminating the need to centralize biometric data. The system achieved 98.36% classification accuracy using VGG19 with FedAvg across distributed clients, demonstrating that privacy-preserving training does not require a performance trade-off in this setting. The work was published in *Engineering Applications of Artificial Intelligence* (Q1, Impact Factor: 8.0) and has received 20 citations since 2024.

Subsequent work extended federated learning into medical imaging privacy-preserving detection of kidney stones, liver lesions, and leukemia from clinical images across distributed hospital settings and surveyed the open challenges facing federated systems in IoT environments. Across these projects, a central question emerged: federated learning addresses data privacy, but it does not address model trustworthiness. A model trained without sharing data can still produce outputs that are unexplainable, hallucinated, or systematically unreliable.

My M.Sc. thesis, BanglaLex, addresses this directly. It is a fine-tuned LLM for legal advisory built with retrieval-augmented generation and explicit hallucination grounding. The domain was chosen because it represents an extreme version of a general problem: a high-stakes application, a low-resource linguistic context, and a near-total absence of existing NLP infrastructure. The grounding techniques developed here are applicable to any low-resource domain where reliability matters. The thesis is currently under evaluation, with expected completion in July 2026.

To date, I have co-authored 5 peer-reviewed journal articles (including 4 in Q1 venues), 1 conference paper, and 3 Springer book chapters, accumulating over 65 citations across

publications. My work spans federated learning, explainable AI, NLP, and large language models.

Teaching and Mentorship

I currently serve as a Lecturer in the Department of Computer Science and Engineering at the University of Asia Pacific, teaching Computer Fundamentals (50+ students) and Competitive Programming (60+ students). I mentor 8+ undergraduate capstone projects in federated learning and NLP applications and previously served as a Teaching Assistant for OOP and DSA Lab, conducting weekly sessions for 60+ students.

Teaching at this level while actively publishing has reinforced my commitment to academic environments — and my belief that the strongest research happens where rigorous inquiry and genuine pedagogy coexist.

Problem-Solving and Analytical Thinking

Alongside my research, I have maintained a strong competitive programming practice that has shaped how I approach technical problems. I have appeared three times as an ICPC Asia Dhaka Regional contestant (2021, 2023, 2024), participated in 10+ national programming contests, and solved over 2,000 problems across Codeforces, CodeChef, AtCoder, and other major online judges. I hold Specialist rank on Codeforces and 3-star rating on CodeChef.

I was awarded the ICPC-Jamilur Reza Choudhury Scholarship three times a merit-based scholarship for top-performing contestants and placed 1st Runner Up and 2nd Runner Up at consecutive EEE Tech Fest Programming Contests.

This background has shaped how I approach research: decompose the problem carefully, identify constraints, find the minimal solution that satisfies them. Competitive programming and research are different disciplines, but the underlying habit of rigorous, structured thinking connects them directly.

Leadership and Service

I served as President of the UAP Programming Contest Club (PCC) from December 2023 to July 2024, leading club operations, organizing intra-university contests, and mentoring members preparing for ICPC. Simultaneously, I served as President of the Robotics Club in the CSE Department, directing workshops and inter-university competitions. I continue to contribute as Co-Convenor of the PCC, coordinating training sessions and contest activities.

In November 2024, I volunteered at the 45th ICPC World Finals, held in Dhaka, Bangladesh the first time the World Finals was hosted in Bangladesh. Contributing to the organization of

the most prestigious collegiate programming contest in the world, on home soil, was a milestone I am proud to carry into my academic profile.

Looking Forward

I am seeking a PhD position to pursue the deeper research questions my work has raised.

The problems I find most compelling sit at the boundary of what current methods can handle: How do we evaluate factual reliability in languages where ground-truth validation corporation does not exist? How must federated aggregation strategies change when client data distributions are linguistically diverse? How do we design explainability mechanisms that are meaningful to users who do not share the assumptions baked into standard interpretability tools?

These are not questions specific to any one language or country. They are open problems in the field. I use low-resource and underserved contexts as my primary testbed because that is where the problems are sharpest and the need is greatest but the answers, if found, will matter everywhere.

I am looking to collaborate with supervisors working on trustworthy AI, federated learning, privacy-preserving systems, or low-resource NLP researchers who are interested in the hard cases, not just the benchmark datasets.

For collaboration or supervision inquiries: [connectplabon@gmail.com] · [[LinkedIn](#)] · [[Google Scholar](#)]